Creating VPC with NAT gateway.

1. What is a VPC

A **Virtual Private Cloud (VPC)** is a logically isolated virtual network in AWS where you can launch AWS resources such as EC2 instances, databases, and load balancers.

It allows you to **control networking components** like IP ranges, subnets, route tables, internet connectivity, and security.

2. Components Required for VPC with NAT Gateway

To build a proper VPC architecture, the following components are used:

Component	Purpose
VPC	Private virtual network in AWS
Subnet	Division of VPC network
Internet Gateway (IGW)	Allows communication between VPC and Internet
NAT Gateway	Allows private subnet instances to access internet
Route Table	Controls network traffic routing
Elastic IP	Static public IP used by NAT Gateway

3. VPC Network Design Example

Example CIDR block:

VPC CIDR : 10.0.0.0/16

Subnets:

Subnet	CIDR	Type
Public Subnet	10.0.1.0/24	Internet access
Private Subnet	10.0.2.0/24	Internal resources

4. Internet Gateway (IGW)

An **Internet Gateway** allows resources in a public subnet to communicate with the internet.

Features:

- Horizontally scalable
- Highly available
- Managed by AWS
- Attached to a VPC

Example route in public route table:

Destination : 0.0.0.0/0
Target : Internet Gateway

5. NAT Gateway

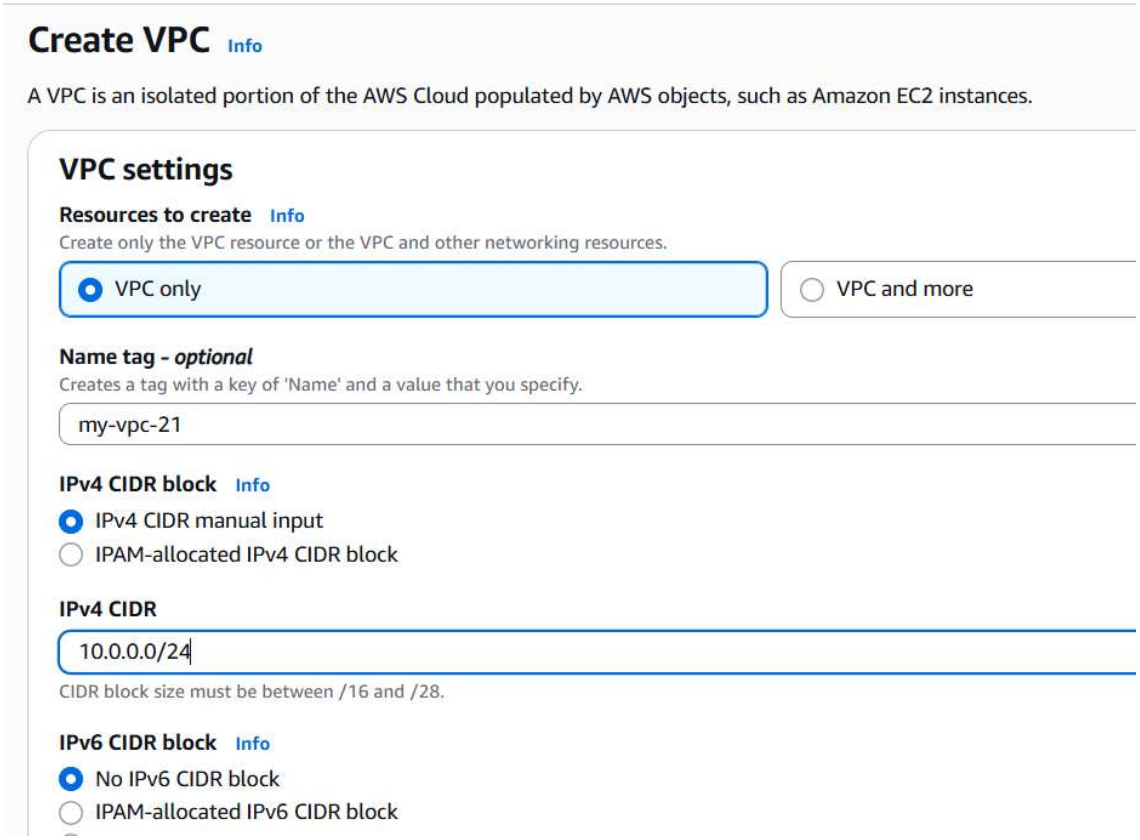
A **NAT Gateway (Network Address Translation Gateway)** allows instances in a **private subnet** to access the internet **without exposing them to incoming internet traffic**.

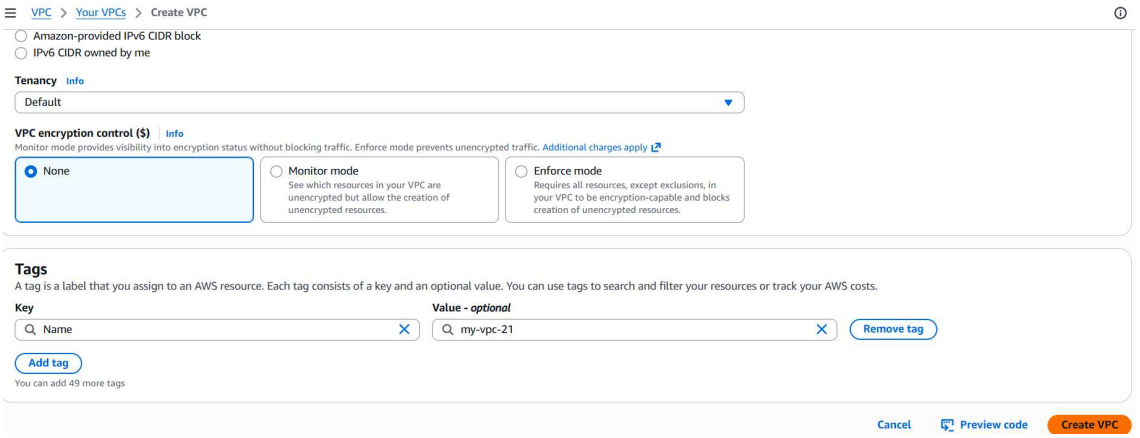
Example:

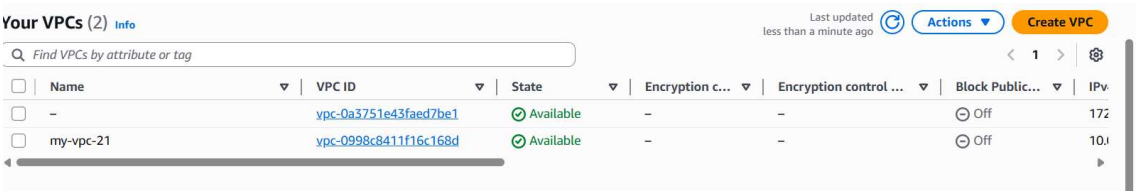
- Private EC2 can download updates
- Install packages
- Access external APIs

But **internet cannot initiate connection to private instances**.

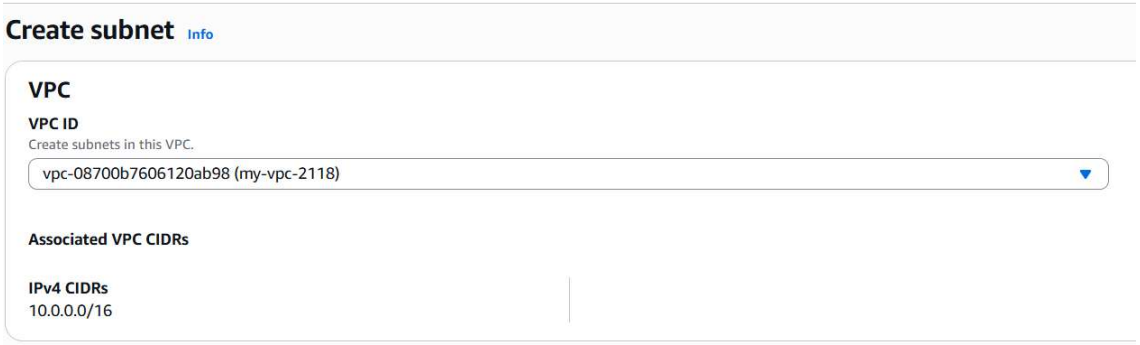
Creating an VPC



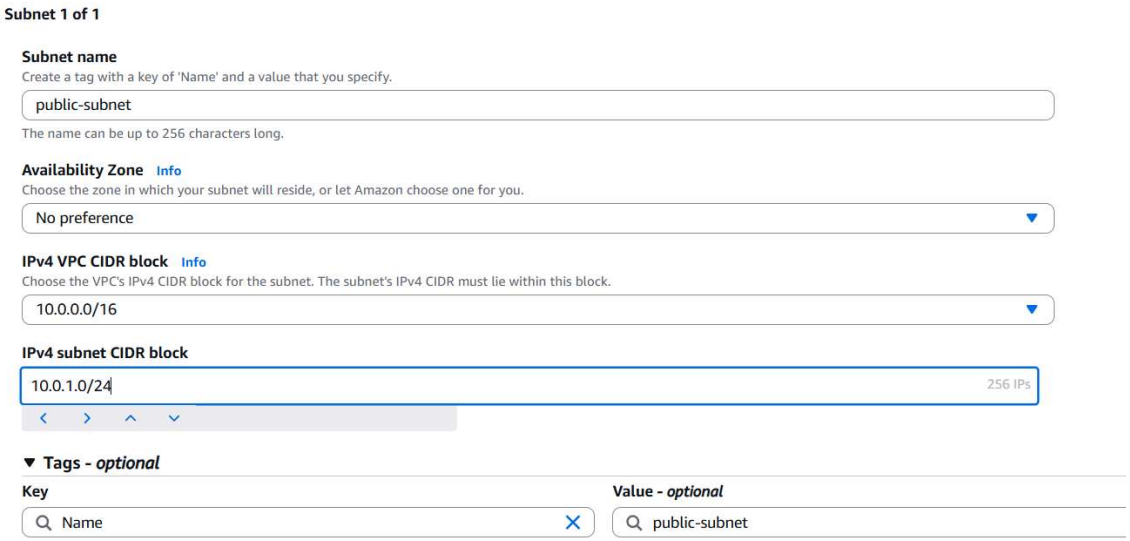




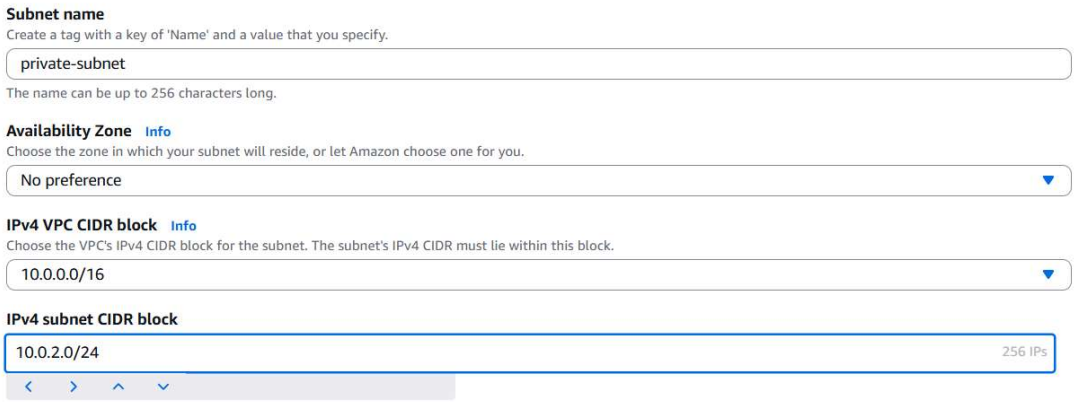
Then we have to create public and private subnet.



For public

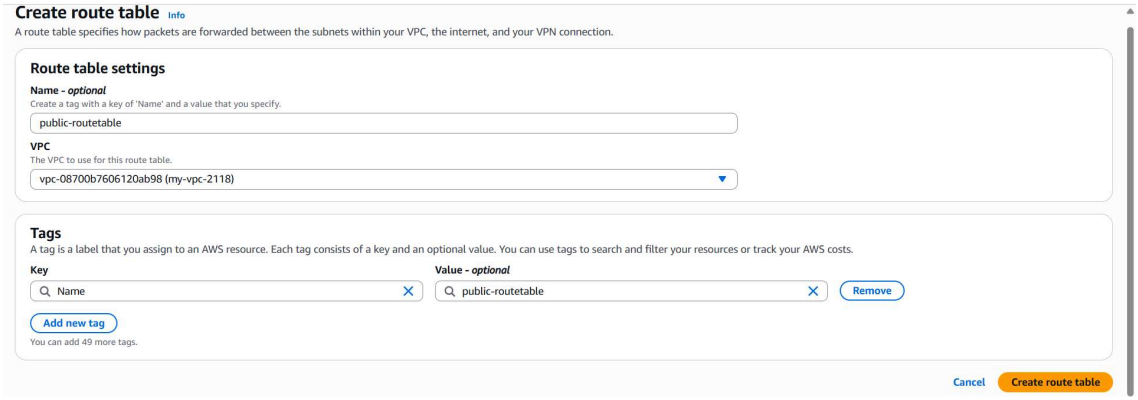


For private

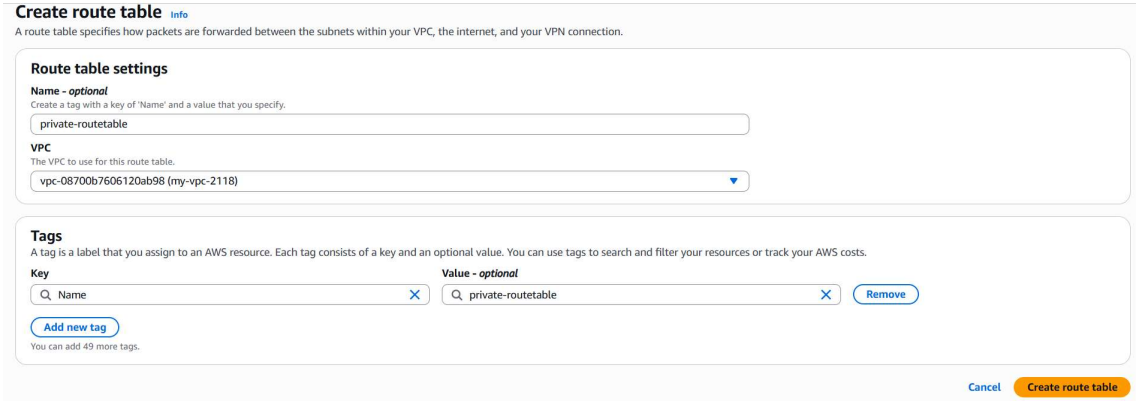


Then we have to create two Route Tables.

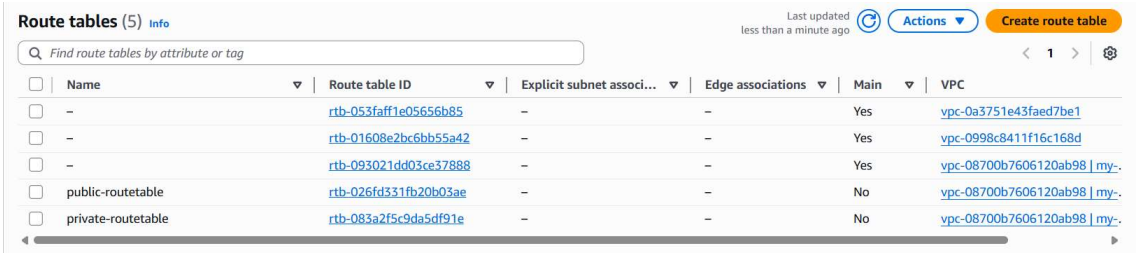
Public route table



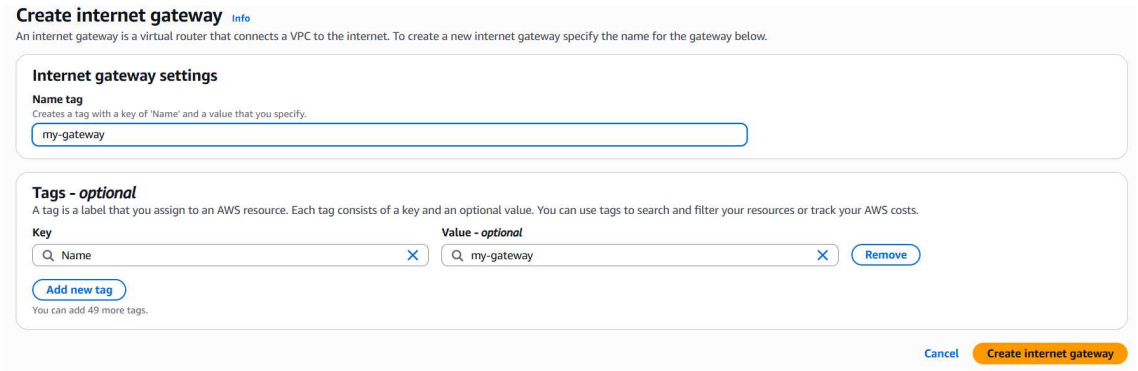
Private route table



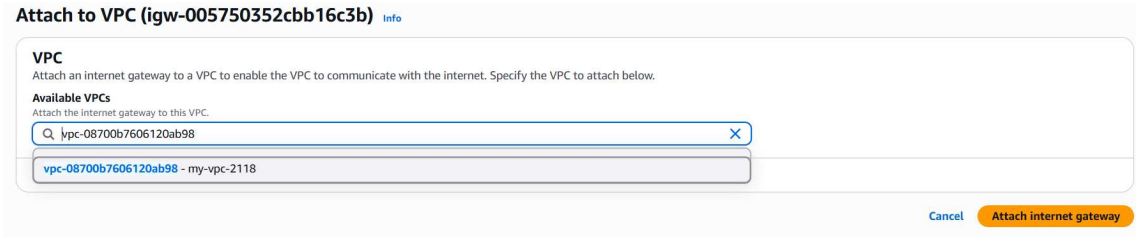
Created two Route Tables.

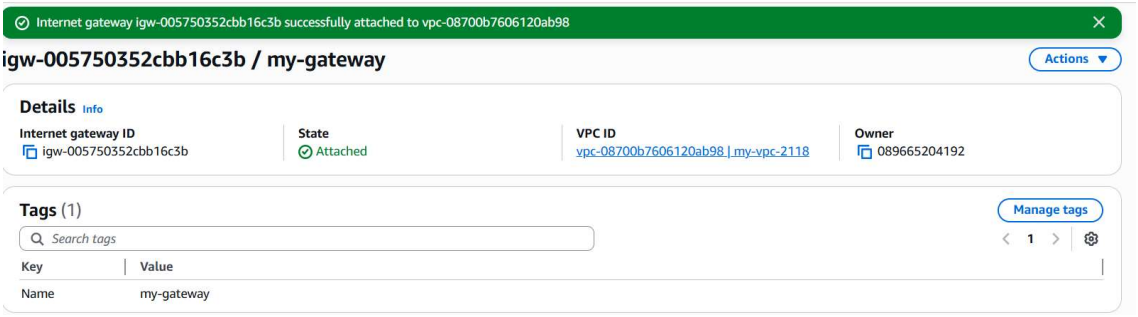


Then we have to create an internet gateway.

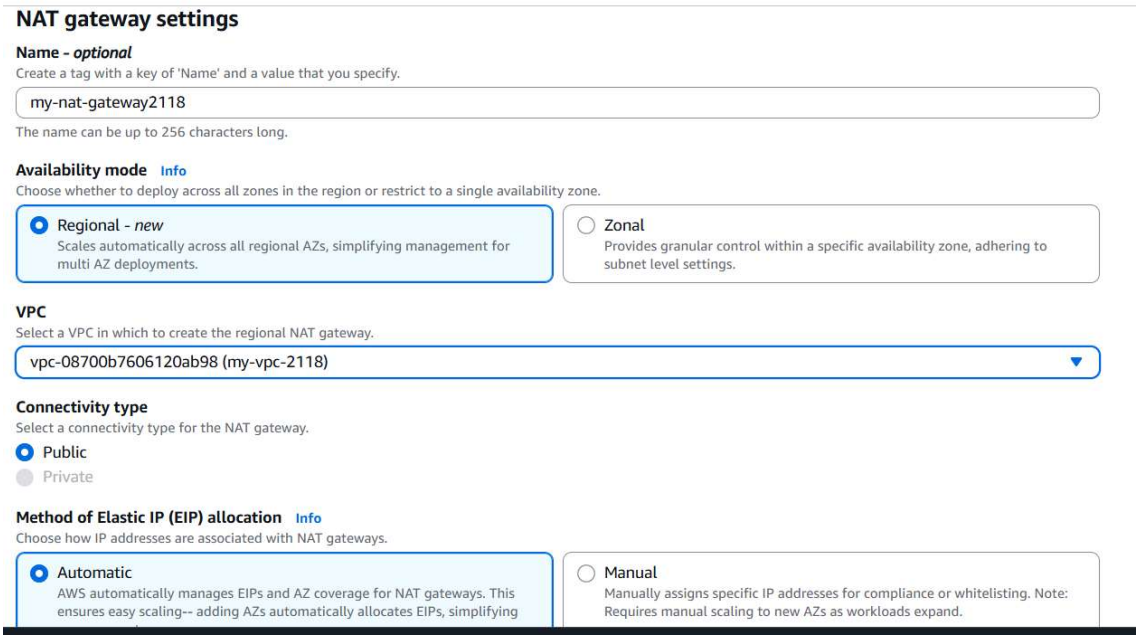


After creation we have to attach to the VPC.

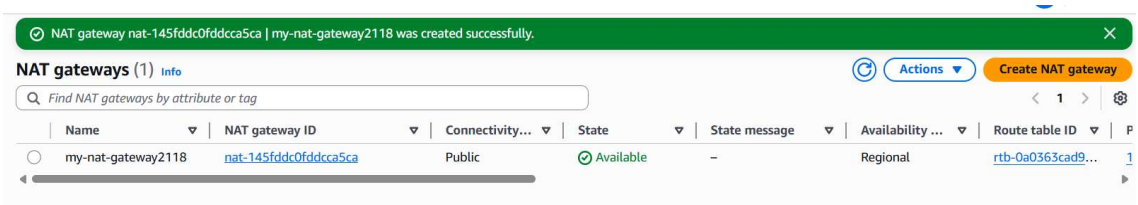




Creation of NAT Gateway.

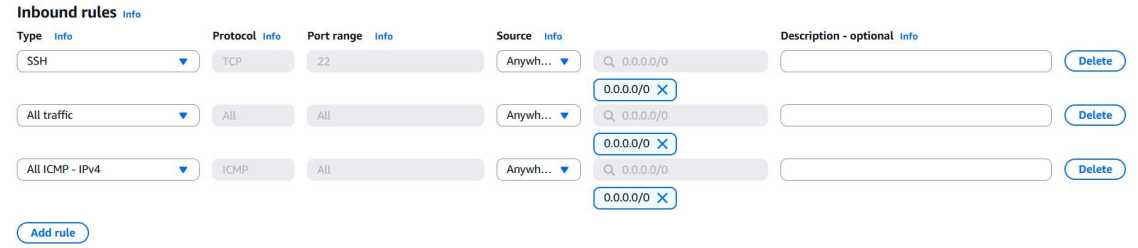


Created NAT gateway.

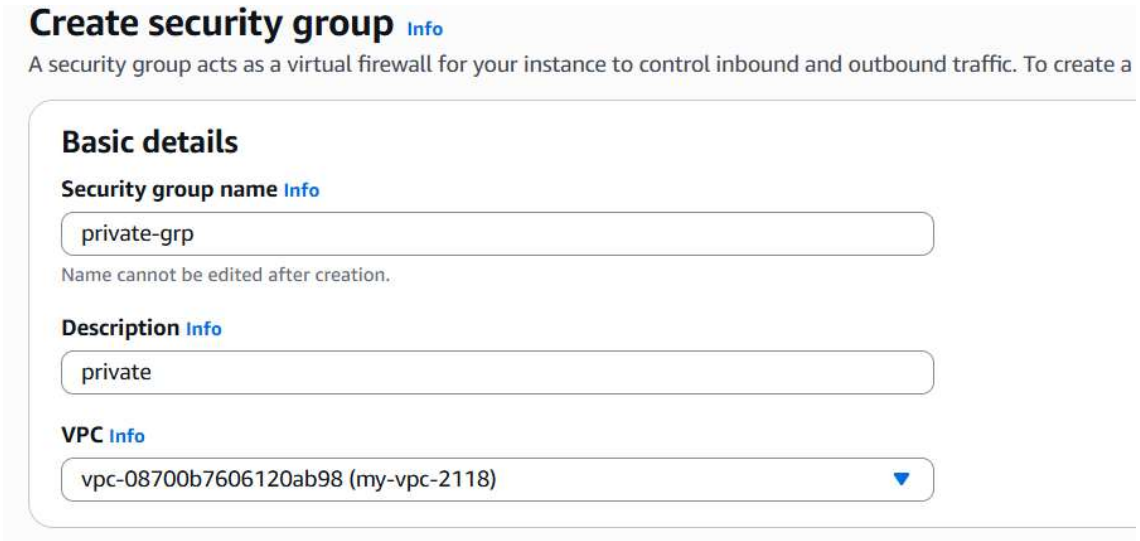


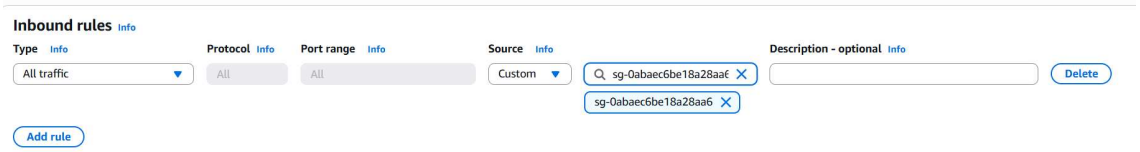
Then we have to create two security groups for the public and private instance.

For public instance inbound rules are.

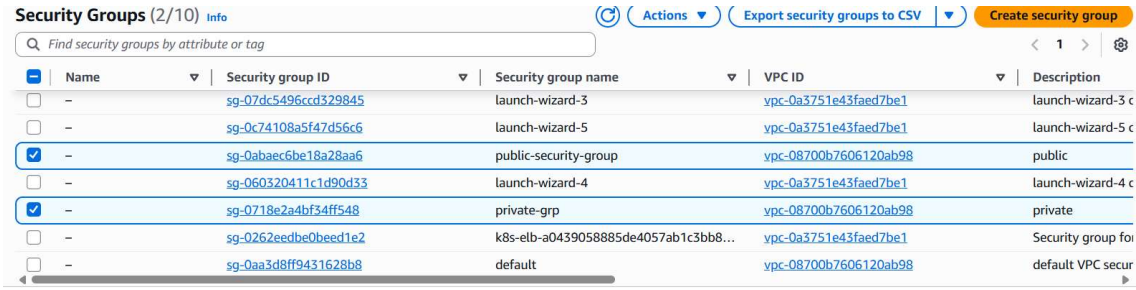


For private instance inbound rules are.



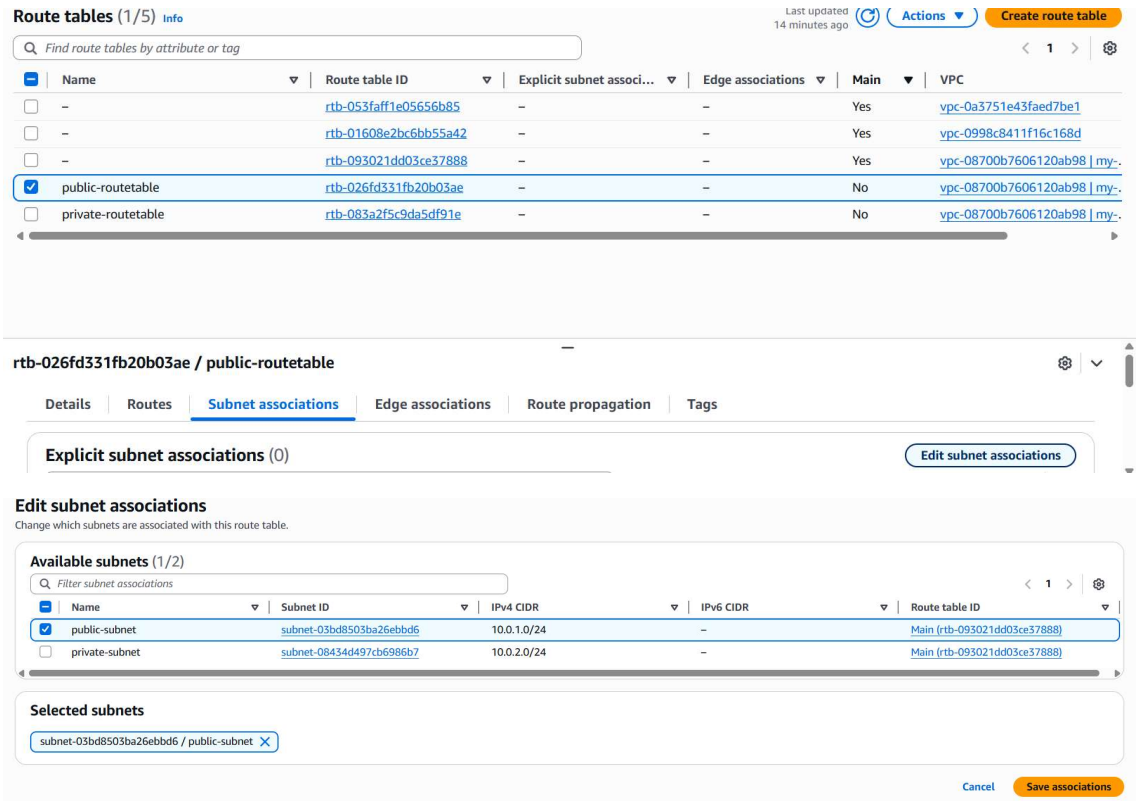


The created security group

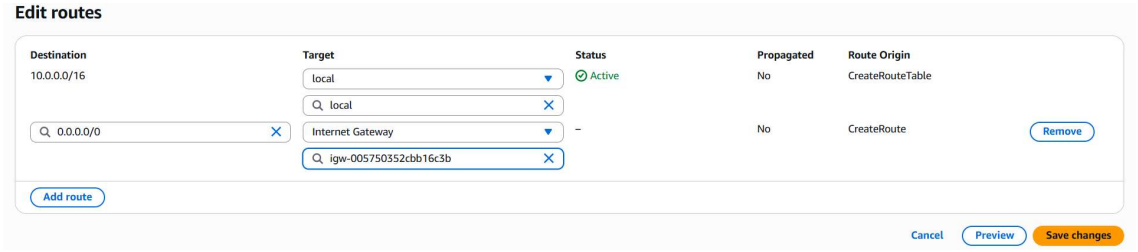


Association of subnets to Route Table.

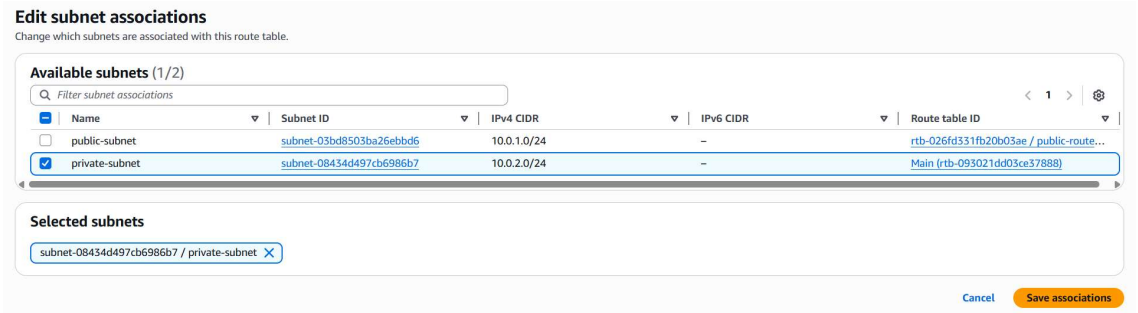
For public Route Table.

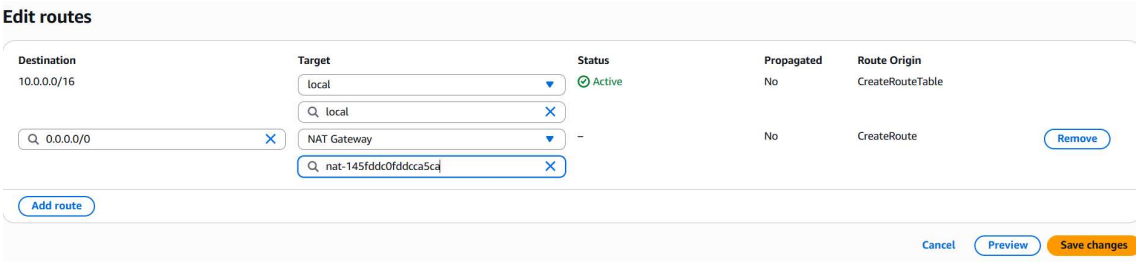


Then Route also



For private Route Table.





Then we have to Launch the public and private instance.

For public instance.

VPC - *required* | Info

vpc-08700b7606120ab98 (my-vpc-2118)
10.0.0.0/16

Subnet | Info

subnet-03bd8503ba26ebbd6 public-subnet
VPC: vpc-08700b7606120ab98 Owner: 089665204192
Availability Zone: ap-south-1c (aps1-az2) Zone type: Availability Zone
IP addresses available: 251 CIDR: 10.0.1.0/24)

Create new subnet

Auto-assign public IP | Info

Enable

Firewall (security groups) | Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

Create security group | Select existing security group

Common security groups | Info

Select security groups

Compare security group rules

public-security-group sg-0abaec6be18a28aa6
VPC: vpc-08700b7606120ab98

Security groups that you add or remove here will be added to or removed from all your network interfaces.

Advanced network configuration

For private instance.

Network settings Info

VPC - *required* | Info

vpc-08700b7606120ab98 (my-vpc-2118)
10.0.0.0/16

Subnet | Info

subnet-08434d497cb6986b7 private-subnet
VPC: vpc-08700b7606120ab98 Owner: 089665204192
Availability Zone: ap-south-1c (aps1-az2) Zone type: Availability Zone
IP addresses available: 251 CIDR: 10.0.2.0/24)

Create new subnet

Auto-assign public IP | Info

Disable

Firewall (security groups) | Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

Create security group | Select existing security group

Common security groups | Info

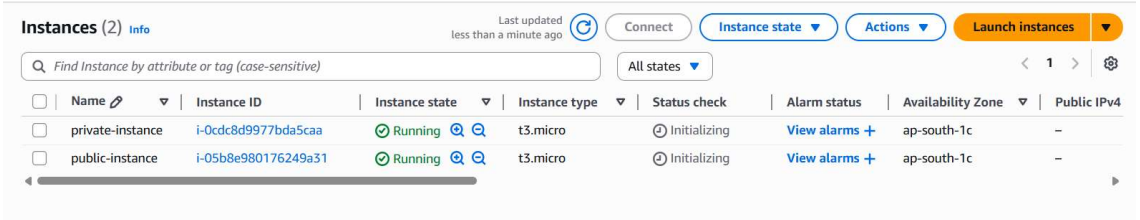
Select security groups

Compare security group rules

private-grp sg-0718e2a4bf34ff548
VPC: vpc-08700b7606120ab98

Security groups that you add or remove here will be added to or removed from all your network interfaces.

Then the created instances.



Then I have checked the network connection using the ping command.

```
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-10-0-1-169 ~]$ sudo su
[root@ip-10-0-1-169 ec2-user]# ping 13.200.207.80
PING 13.200.207.80 (13.200.207.80) 56(84) bytes of data.
64 bytes from 13.200.207.80: icmp_seq=1 ttl=126 time=0.245 ms
64 bytes from 13.200.207.80: icmp_seq=2 ttl=126 time=0.238 ms
64 bytes from 13.200.207.80: icmp_seq=3 ttl=126 time=0.215 ms
64 bytes from 13.200.207.80: icmp_seq=4 ttl=126 time=0.248 ms
64 bytes from 13.200.207.80: icmp_seq=5 ttl=126 time=0.237 ms
64 bytes from 13.200.207.80: icmp_seq=6 ttl=126 time=0.228 ms
64 bytes from 13.200.207.80: icmp_seq=7 ttl=126 time=0.229 ms
64 bytes from 13.200.207.80: icmp_seq=8 ttl=126 time=0.225 ms
^C
--- 13.200.207.80 ping statistics ---
8 packets transmitted, 8 received, 0% packet loss, time 7289ms
rtt min/avg/max/mdev = 0.215/0.233/0.248/0.010 ms
[root@ip-10-0-1-169 ec2-user]# ping 13.200.207.80
```

Then I have connected to the private instance through the public instance with the help of the private ip of the private instance.

```
[ec2-user@ip-10-0-1-213 ~]$ nano privatekey
[ec2-user@ip-10-0-1-213 ~]$ chmod 400 privatekey
[ec2-user@ip-10-0-1-213 ~]$ ssh -i "privatekey" ec2-user@10.0.2.52
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-10-0-2-52 ~]$
```

Then I have checked the network connection in the private instance also.

```
[ec2-user@ip-10-0-2-137 ~]$
[ec2-user@ip-10-0-2-137 ~]$ ping 10.0.2.137
PING 10.0.2.137 (10.0.2.137) 56(84) bytes of data.
64 bytes from 10.0.2.137: icmp_seq=1 ttl=127 time=0.023 ms
64 bytes from 10.0.2.137: icmp_seq=2 ttl=127 time=0.036 ms
64 bytes from 10.0.2.137: icmp_seq=3 ttl=127 time=0.030 ms
64 bytes from 10.0.2.137: icmp_seq=4 ttl=127 time=0.029 ms
64 bytes from 10.0.2.137: icmp_seq=5 ttl=127 time=0.031 ms
^C
--- 10.0.2.137 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4158ms
rtt min/avg/max/mdev = 0.023/0.029/0.036/0.004 ms
[ec2-user@ip-10-0-2-137 ~]$
```

The instances have been created in the custom VPC.